X PENSE TRACKER

Every Rupee Accounted For

# Project Development Document

## Complete Prompt Log and Build History

| | |
|---|---|
| **Project** | X-Pense Tracker — personal expense tracking application |
| **Author** | Senthil Kumar Packiyam |
| **Document** | Prompt log, requirement trail and delivery record |
| **Current version** | XPENSE-V2 / XPENSE-V2-PWA (build 2026-08-12) |
| **Storage schema** | localStorage key expense-tracker-v1, schema v4 |
| **Deployment** | https://leafy-travesseiro-f06ff8.netlify.app |
| **Prompts logged** | 19 across 11 development phases |
| **Date issued** | 7 October 2026 |

# Contents

*Right-click the field below and choose Update Field to populate page numbers.*

# 1. Purpose and Scope

This document records the complete development of X-Pense Tracker as a sequence of specification prompts and the work delivered against each one. It serves three purposes: as a requirements trail showing how the product was specified, as a change log showing what each revision altered, and as a record of the design and deployment decisions taken along the way.

Every prompt is reproduced verbatim in Section 3, in the order it was issued, each paired with a summary of what was delivered in response. Section 2 gives the product overview, Section 4 the version history, and Section 5 the technical architecture.

## 1.1 Product summary

X-Pense Tracker is a personal expense tracking application for the Indian market, built as a single self-contained HTML file with no backend, no external dependencies and no user account. All data is held in the browser's local storage on the user's own device.

- Entry capture — amount, date, two-level category, notes, payment mode and bank account
- Bank balance engine — manual balance setting, with automatic adjustment only for payment modes that genuinely move money through a bank account
- Categorisation — 14 parent categories, 60+ sub-categories, with keyword-driven auto-suggestion from the notes field
- Summary — donut and trend charts across daily, weekly, monthly, quarterly and yearly ranges, with breakdown by category, sub-category or payment mode
- Transaction log — full table view with category, sub-category, mode and period filters plus global keyword search
- Portability — JSON backup and restore, CSV export for Excel

## 1.2 Design principles

Three constraints were held throughout and shaped most of the technical decisions:

- Self-contained — one HTML file, assets embedded as base64, no CDN calls, so the app runs from a file manager, a web server or an installed app shell without modification
- Privacy by architecture — no server means no data transmission; the privacy policy states that nothing is collected because nothing can be
- Extensible by structure — reference data (categories, keywords, modes, cards) is declared as plain arrays at the top of the script, so extending the taxonomy is a data edit rather than a code change

## 2. How to Read the Prompt Log

Each entry in Section 3 follows the same structure:

- Prompt — the instruction exactly as issued, preserving original wording, spelling and formatting. No edits have been made.
- Attachments — any image supplied with the prompt, where applicable.
- Delivered — a summary of the implementation that followed, including schema changes and version impact.

Prompts are grouped into eleven development phases, from the initial build through brand identity, taxonomy expansion, visual refinement, deployment and distribution. The phase name appears above each prompt.

Verbatim prompt text appears in a bordered block in monospaced type. Where a prompt contained a long bulleted specification, the structure of that list has been preserved.

# 3. Prompt Log

## Phase 1 — Core build

### Prompt 1 — Initial specification

*Attachment: Reference screenshot of an iOS expense-tracker UI (5 screens)*

```
Build a single-page HTML web app for personal expense tracking with local persistent
storage (JSON file-based, retaining data across sessions). Include the following
  features:

1. Data Entry
* Add daily expense entries with: amount, date, category (manual free-text entry, with
  autosuggest from a predefined global category list), description/notes, and mode of
    payment.
* Mode of payment options: UPI, Credit Card, Rupay Credit Card, Cash, Netbanking,
  ATM Withdrawal.
* If mode is UPI, prompt for the linked bank account.
* Add an option to log credited/incoming money, which adds to the current bank
  balance.
* Bank Balance Logic
* Let me set/edit the current bank balance manually.
* Auto-deduct from bank balance only for: UPI, Netbanking, ATM Withdrawal.
* Do NOT deduct from bank balance for: Credit Card, Rupay Credit Card, Cash.
* Credited amounts should add back to the bank balance.
* Categorization
* Auto-suggest a category based on keywords in the entered description/details (basic
  keyword-matching logic is fine).
* Provide a comprehensive predefined list of global categories, including: Investments
  (SIP, LIC, Trading/Stocks, Mutual Funds, PPF, etc.), EMI, Health/Medical, Household,
  Groceries, Utilities, Rent, Renewals/Subscriptions, Travel, Dining, Entertainment,
  Education, Insurance, Shopping, Fuel/Transport, and any other commonly used
    categories.
* Include an "Others" category with a free-text field to specify details when nothing
  else fits.
* Dashboard
* Show total expenses, current bank balance, and a breakdown by category.
* Add time-range filters: Daily, Weekly, Monthly, Quarterly, Yearly — supporting
  historical data up to 2 years back.
* Use simple charts/visuals (bar/pie) where useful.
* Transaction Log / Table View
* Display all entries in a row/table format (date, amount, category, mode, bank,
  notes).
* Add filtering by category.
* Add a global search bar that searches across all columns/fields for a keyword.
* Storage
* Persist all data locally in JSON (or XML if more practical) so that closing and
  reopening the app retrieves the last saved state — no external database or server
    required.
* Design
```

```
* Clean, minimal UI with a plain white background.
* Use a simple, professional color accent scheme as same as attached one for your
  review.

Please build this as a self-contained HTML/CSS/JavaScript app (using localStorage or a
local JSON file for persistence), organized clearly so I can extend it later (e.g.
  export
to Excel, add budgets/alerts).
```

**Delivered**

Single-file HTML app delivered: 39 categories, 6 payment modes, bank-balance engine, SVG donut and trend charts, searchable transaction table, localStorage persistence (schema v1), CSV/JSON export.

## Phase 2 — Brand identity

### Prompt 2 — Logo and colour system

*Attachment: X-PENSE TRACKER logo (PNG)*

```
Two changes i want to do:

1. Add the attached logo with the app
2. Make all the letters with the combination of the logo color attached.
3. make all the font as same format but the size can vary
```

**Delivered**

Logo background removed, quantised and embedded as a base64 data URI. Palette resampled from the logo itself (#00B050 green, #0070C0 blue); all 39 category colours remapped onto a green-to-blue arc. Single type family enforced, weight capped at 700.

## Phase 3 — Taxonomy

### Prompt 3 — Two-level category structure

```
Add the below categories and sub categories

Housing — Rent or home loan EMI; Maintenance / society charges; Property tax;
  Repairs, furnishing, appliances
Utilities — Electricity (BESCOM), water, piped gas / LPG cylinder; Broadband, mobile
  recharges, DTH; Cook, maid, driver, security
Food & Groceries — Kirana / supermarket / online grocery (BigBasket, Blinkit, Zepto);
  Milk, vegetables, fruits; Eating out and food delivery (Swiggy, Zomato)
Transport — Fuel, FASTag / tolls, parking; Cab and auto (Ola, Uber, Rapido), metro/bus
  passes; Vehicle insurance, servicing, EMI
Health — Health insurance premiums; Doctor consultations, medicines, diagnostics;
  Dental, eye care, gym / fitness
Education — School / college fees, tuition and coaching; Books, uniforms, stationery,
  school transport; Courses and certifications
Insurance & Protection — Term life, health top-up, personal accident; Home and vehicle
  insurance
```

```
Investments & Savings (technically outflow, not expense — but worth tracking) —
  SIPs, PPF/EPF/NPS, RD/FD; Direct equity, ETFs, gold
Debt Repayment — Home / vehicle / personal loan EMIs; Credit card bill payments
Taxes — Income tax (advance tax, self-assessment); Professional tax, GST on business
  income
Lifestyle & Discretionary — Clothing, footwear, personal care, salon; Subscriptions
  (OTT, cloud, apps); Electronics and gadgets; Hobbies, books, sports
Family & Social — Festivals (Diwali, Pongal, etc.), gifts; Weddings, functions,
  contributions to relatives; Donations, temple offerings; Support to parents /
    dependents
Travel — Domestic and international trips; Hometown visits, flights/trains, hotels,
  visas, forex
Miscellaneous — Bank charges, ATM fees, late fees; Legal, CA and documentation fees;
  Emergency / unplanned
```

**Delivered**

Data model restructured to category + sub-category (schema v2). 14 parent categories, 60+ sub-categories, 7 income categories. ~150 keyword rules now resolve to both levels. Existing v1 entries auto-migrated via a mapping table. Summary gained a category/sub-category toggle.

## Phase 4 — Payment modes

### Prompt 4 — Payment mode revision

```
in payment mode section do the below changes:

1. remove Rupay credit card option
2. add Auto-debit / mandates
3. add Cheque/DD
4. Other and create an option as "Details" to manually enter the details of the mode
   of the payment made.
```

**Delivered**

Modes reworked (schema v3). Auto-debit/Mandate and Cheque/DD added as bank-affecting modes. "Other" opens a required free-text details field plus a checkbox controlling whether the entry touches the bank balance. Legacy Rupay entries migrated to Credit Card with the detail preserved.

## Phase 5 — Visual refinement

### Prompt 5 — Pure white surfaces

```
change the background color as pure white. no more shadow colors
```

**Delivered**

All tinted panels and drop shadows removed. Structure now carried by hairline borders; active states use solid fills. No rgba() values remain in the stylesheet.

### Prompt 6 — Typeface change

*Attachment: Screenshot of the "Add expense" heading*

> change the attached "add expense" font into "TW cen MT" and no italic

**Delivered**

Tw Cen MT set as the family across the whole app with a Century Gothic / Futura fallback stack. Every italic rule removed.

### Prompt 7 — Type scale increase

*Attachment: Screenshot of the Add expense screen*

> change the size of the font little bigger

**Delivered**

Every font size scaled ~13% proportionally (body 15→17px, titles 34→38.5px). Containers widened to match.

## Phase 6 — Version control

### Prompt 8 — Snapshot tagging

> save this final output and tag it. when i recall with the tag in future, make sure to bring this snapshot as same. let me know the tag to use in future.

**Delivered**

Tag XPENSE-V1 established and stamped inside the file (HTML comment plus the Settings → Version row). Snapshot archived as a dated filename.

## Phase 7 — Mobile strategy

### Prompt 9 — Mobile app feasibility

> Is there a way to launch this app as mobile app for android and IOS phones

**Delivered**

Three routes assessed: PWA (recommended), PWABuilder APK wrapper, and Capacitor native builds. Per-device storage and iOS cache-eviction risks flagged.

### Prompt 10 — Monetisation assessment

> lets focus on only android phones as of now. Would this be monetized even if the users download it as mobile app and use it for free?

**Delivered**

Analysis of freemium, one-time unlock, ads and affiliate models against the TWA vs Capacitor wrapper constraint. Noted that paid features require a backend, and that Play Console personal accounts face a 12-tester / 14-day closed-test gate.

# Phase 8 — Deployment

## Prompt 11 — Start the free release

> ok, I will ship free and unmonetize it first. Let me know where do we start?

**Delivered**

PWA package built: manifest, service worker with offline caching, five-size icon set generated from the logo mark, safe-area and small-screen CSS, home-screen shortcuts. Tagged XPENSE-V1-PWA. DEPLOY.md written.

## Prompt 12 — Hosting — locating the site URL

> i dropped xpense-pwa.zip file and it is dropped and i am able to open. However, I am not
> able to find the https url, where do i get it from the website. guide me step by step from basic

**Delivered**

Root cause identified: the archive wrapped files in a subfolder, so index.html did not land at the site root. Repackaged with files at the archive root. Step-by-step Netlify Drop walkthrough provided.

## Prompt 13 — URL not visible on dashboard

*Attachment: Screenshot of the Netlify project overview*

> I dont see any URL once i dropped (attached screenshot for your reference

**Delivered**

URL located on the dashboard. Site was flagged Private, which would have blocked mobile access and service-worker registration — "Make public" identified as the required action.

## Prompt 14 — Deployment confirmed

> i am able to launch and open the url now

**Delivered**

Live at https://leafy-travesseiro-f06ff8.netlify.app. Verification steps and Android install instructions provided.

# Phase 9 — Distribution

## Prompt 15 — Play Store submission

> i want to add this app into google playstore

**Delivered**

Store assets produced: 512×512 listing icon, 1024×500 feature graphic, full privacy policy page, store listing copy. Six-phase submission plan covering Play Console registration, PWABuilder packaging, assetlinks.json digital asset links, declarations, and the closed-testing requirement.

### Prompt 16 — Web deployment clarification

> is there a way i can deploy it as web page ?

**Delivered**

Confirmed the existing Netlify URL already is the web deployment. Custom domain and site renaming options outlined; Play Store positioned as discovery-only, not a prerequisite.

### Prompt 17 — Version recall

> could you recall the x-pense tracker app the latest version?

**Delivered**

XPENSE-V1-PWA identified as current. Difference between the standalone and PWA builds documented; outstanding privacy-policy placeholder flagged.

## Phase 10 — Card tracking

### Prompt 18 — Credit card sub-modes and payment-mode breakdown

> I want to add two modifications:
>
> 1. when i select the credit card, add the sub payment mode with different credit card as
>  Scapia credit card, HDFC credit card, and ICICI credit card.
> 2. once i enter these values, i want to see the "summary" page with the "breakdown" section by payment mode which should contain, all the above credit card wise and other
> payment mode show under "Bank balance". Basically I should see 4 different payment mode
> in summary --> Breakdown. Scapia credit card, HDFC credit card, and ICICI credit card,
> and Bank balance. Further subdivide the bank balance based on the Bank account that i
> add it under "settings" page.

**Delivered**

Schema v4. Card selection required when Credit Card is chosen; cards manageable in Settings. Summary gained a third breakdown level, "By payment mode", with one slice per card and a Bank balance bucket subdivided by account. Cash and non-bank "Other" entries isolated into a separate bucket to keep the bank figure accurate. Grouping verified against a reconciliation test.

# Phase 11 — Documentation

## Prompt 19 — Project documentation

> get me the downloadable copy of the above all prompts to submit as project document.

**Delivered**

This document.

# 4. Version History

Each stable milestone was tagged and the tag stamped inside the file itself — as an HTML comment below the title and in the Settings → Version row — so any build can identify itself.

| Tag | Build date | Changes |
|---|---|---|
| XPENSE-V1 | 4 Aug 2026 | First tagged standalone build. Schema v3. Logo, palette, two-level taxonomy, revised payment modes, white surfaces, Tw Cen MT at the larger type scale. |
| XPENSE-V1-PWA | 5 Aug 2026 | Progressive Web App layer: manifest, service worker (cache xpense-v2), icon set, safe-area CSS, home-screen shortcuts, privacy policy page. |
| XPENSE-V2 | 12 Aug 2026 | Credit card sub-modes with per-card tracking. Summary breakdown by payment mode with bank balance subdivided by account. Schema v4. |
| XPENSE-V2-PWA | 12 Aug 2026 | As above, with the PWA layer. Service worker cache bumped to xpense-v3. |

## 4.1 Storage schema evolution

The storage key has remained expense-tracker-v1 throughout. The schema version inside it has advanced four times, each with an automatic migration that runs on load so no data is lost between versions.

| Schema | Introduced | Migration performed |
|---|---|---|
| v1 | Prompt 1 | Initial structure — flat category per entry. |
| v2 | Prompt 3 | Flat category split into category + sub-category via a mapping table; unmapped values preserved in the details field. |
| v3 | Prompt 4 | Rupay Credit Card folded into Credit Card with the card named in payment details; Cheque renamed to Cheque / DD. |
| v4 | Prompt 18 | Credit card register added; card field introduced on entries. Pre-existing card entries show as Unassigned card until edited. |

# 5. Technical Reference

## 5.1 Payment modes and bank balance logic

The central rule of the application is that only payment modes which genuinely move money through a bank account adjust the tracked balance. This was specified in Prompt 1 and extended in Prompts 4 and 18.

| Mode | Affects balance | Notes |
|---|---|---|
| UPI | Yes — deducts | Prompts for the linked bank account. |
| Netbanking | Yes — deducts | Prompts for the account used. |
| ATM Withdrawal | Yes — deducts | Prompts for the account used. |
| Auto-debit / Mandate | Yes — deducts | Added in Prompt 4. |
| Cheque / DD | Yes — deducts | Added in Prompt 4. |
| Credit Card | No | Settled separately. Requires card selection (Prompt 18). |
| Cash | No | Cash in hand. |
| Other | Conditional | Free-text details required; a checkbox controls whether it moves the balance. |

Incoming money follows the mirror rule: credits received through a bank mode increase the balance; cash received does not.

## 5.2 Balance calculation

Each account stores a manually set baseline and the timestamp at which it was set. The displayed balance is that baseline plus every balance-affecting entry recorded after that timestamp. Re-setting a balance establishes a new baseline, so a manual correction is never double-counted against entries that preceded it.

## 5.3 Category taxonomy

Fourteen parent categories, each with its own sub-categories, plus an "Other" sub-category on every parent that opens a free-text field:

- Housing
- Utilities
- Food & Groceries
- Transport
- Health
- Education
- Insurance & Protection
- Investments & Savings
- Debt Repayment

- Taxes
- Lifestyle & Discretionary
- Family & Social
- Travel
- Miscellaneous

Seven income categories cover salary, business and freelance income, investment income, rent received, refunds and cashback, gifts received and other income.

Investments & Savings is flagged in the data model as an outflow rather than consumption, reflecting the note in Prompt 3. When a period contains such entries, the summary states the portion of total spend that went to investments.

## 5.4 Deployment architecture

| Layer | Technology | Purpose |
|---|---|---|
| Application | Single HTML file | Markup, styles, logic and base64 logo in one document. |
| Persistence | localStorage | Key expense-tracker-v1, schema v4. Per-device, no sync. |
| App shell | Web App Manifest | Name, icons, standalone display, theme colour, shortcuts. |
| Offline | Service worker | Network-first for the page, cache-first for assets. |
| Hosting | Netlify | Static hosting over HTTPS, drag-and-drop deployment. |
| Android packaging | PWABuilder | Generates a signed .aab plus assetlinks.json. |

## 5.5 Known constraints

- Data is per-device. The same user on a phone and a laptop maintains two separate ledgers; JSON export and import is the only bridge.
- Clearing browser site data erases all records. Periodic JSON export is the only recovery path.
- Tw Cen MT ships with Microsoft Office. On devices without it the stack falls back to Century Gothic, then Futura, then Helvetica.
- Entries recorded before schema v4 carry no credit card attribution and appear as "Unassigned card" in the payment-mode breakdown.

# 6. Outstanding Items

Items identified during development and not yet closed at the date of this document:

| Item | Raised at | Action required |
|---|---|---|
| Privacy policy placeholder | Prompt 15 | Replace REPLACE-WITH-YOUR-EMAIL with a working contact address before store submission. |
| Play Console registration | Prompt 15 | One-time $25 fee and identity verification. Required before the tester clock can start. |
| Closed testing | Prompt 15 | 12 testers opted in for 14 continuous days before production access unlocks. |
| Digital asset links | Prompt 15 | Serve assetlinks.json at /.well-known/ so the installed app opens without a browser address bar. |
| Signing key backup | Prompt 15 | Store the generated key permanently; losing it prevents all future updates. |
| Store screenshots | Prompt 15 | Two to eight phone screenshots with representative sample data. |

## 6.1 Possible future development

- Budgets and overspend alerts, comparing period totals against limits per category
- Native Excel (.xlsx) export in place of CSV
- Cloud sync to resolve the per-device storage limitation
- Receipt image capture attached to entries
- Recurring-entry templates for rent, EMIs and subscriptions

*End of document*